

Fixed Sticky Refreshing Preserves the Exponent-Two Degree Laws in Range-Renewal Networks

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Abstract

Xie and Zhou proved exact directed and undirected degree laws for the simple graph generated by consecutive independent samples from a heavy-tailed countable distribution. We resolve a fixed dependent specialization. At each clock step, retain the present label with probability $\rho \in [0, 1)$ and otherwise redraw independently from the same distribution; delete self-loops and suppress repeated edges. Pathwise, the resulting graph through clock time n is exactly the independent graph on the labels exposed at the $M_n + 1$ refresh-skeleton positions, where M_n counts iid Bernoulli($1 - \rho$) refreshes. Since $M_n \rightarrow \infty$ almost surely, all four fixed-degree limits transfer along this random index. With observed-range normalization, no factor depending on ρ remains. Taking the degree limit only after the clock-time limit, the directed and undirected tails have amplitudes 1 and 2^γ , and the corresponding local masses have regular-variation exponent -2 . The result is pointwise in fixed $\rho < 1$ and fixed degree before the outer limit; it gives neither a growing-degree limit nor uniformity as $\rho \uparrow 1$.

1 Setting and result

Let $\pi = (\pi_i)_{i \geq 1}$ be a nonincreasing strictly positive probability mass function. Fix $0 < \gamma < 1$ and assume $\pi_k \in \text{RV}_{-1/\gamma}$, meaning that for each $t > 0$,

$$\frac{\pi_{\lceil tx \rceil}}{\pi_{\lceil x \rceil}} \longrightarrow t^{-1/\gamma} \quad (x \rightarrow \infty).$$

For a fixed $\rho \in [0, 1)$, set $q = 1 - \rho > 0$. Start with $X_0 \sim \pi$ and, independently at every clock step, retain X_{t-1} with probability ρ or draw X_t afresh from π with probability q . Equivalently,

$$\mathbb{P}(X_t = y \mid X_{t-1} = x) = \rho \mathbf{1}_{\{x=y\}} + q\pi_y.$$

For $n \geq 0$, put $V_n = \{X_0, \dots, X_n\}$ and $R_n = |V_n|$. The loop-deleted directed and undirected simple edge sets are

$$\begin{aligned} E_n^{\rightarrow} &= \{(X_{t-1}, X_t) : 1 \leq t \leq n, X_{t-1} \neq X_t\}, \\ E_n^{\text{u}} &= \{\{X_{t-1}, X_t\} : 1 \leq t \leq n, X_{t-1} \neq X_t\}. \end{aligned}$$

Because these are sets, repeated edges are suppressed. Write $D_n^{\rightarrow}(x)$ for out-degree and $D_n^{\text{u}}(x)$ for undirected degree. For $a \in \{\rightarrow, \text{u}\}$ and $k \geq 1$, define

$$r_{n,k}^{\rho,a} = \frac{1}{R_n} \#\{x \in V_n : D_n^a(x) = k\}, \quad r_{n,k+}^{\rho,a} = \frac{1}{R_n} \#\{x \in V_n : D_n^a(x) \geq k\}.$$

We recall the deterministic limits from the independent theorem of Xie and Zhou [1, Theorem 1 and Proposition 2]. If K_m denotes the number of distinct values in m independent draws from π , let

$$s_\ell(\gamma) = \frac{\gamma \Gamma(\ell - \gamma)}{\Gamma(1 - \gamma) \Gamma(\ell + 1)},$$

and set

$$r_k^{\rightarrow} = \sum_{\ell \geq 1} s_\ell(\gamma) \mathbb{P}(K_\ell = k), \quad r_k^u = \sum_{\ell \geq 1} s_\ell(\gamma) \mathbb{P}(K_{2\ell} = k), \quad r_{k+}^a = \sum_{j \geq k} r_j^a.$$

Their loop-deleted independent result states that the four observed-range statistics converge almost surely for each fixed k , and that

$$r_{k+}^{\rightarrow} \sim \pi_k^\gamma, \quad r_{k+}^u \sim 2^\gamma \pi_k^\gamma, \quad r_k^{\rightarrow} \sim \frac{\pi_k^\gamma}{k}, \quad r_k^u \sim \frac{2^\gamma \pi_k^\gamma}{k}. \quad (1)$$

Theorem 1. *For each fixed $\rho \in [0, 1)$, almost surely and simultaneously for every fixed integer $k \geq 1$,*

$$r_{n,k}^{\rho,a} \longrightarrow r_k^a, \quad r_{n,k+}^{\rho,a} \longrightarrow r_{k+}^a \quad (n \rightarrow \infty), \quad a \in \{\rightarrow, u\}.$$

Only after these clock-time limits are taken, the four degree asymptotics are

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{\lim_{n \rightarrow \infty} r_{n,k+}^{\rho, \rightarrow}}{\pi_k^\gamma} &= 1, & \lim_{k \rightarrow \infty} \frac{\lim_{n \rightarrow \infty} r_{n,k+}^{\rho, u}}{2^\gamma \pi_k^\gamma} &= 1, \\ \lim_{k \rightarrow \infty} \frac{\lim_{n \rightarrow \infty} r_{n,k}^{\rho, \rightarrow}}{\pi_k^\gamma / k} &= 1, & \lim_{k \rightarrow \infty} \frac{\lim_{n \rightarrow \infty} r_{n,k}^{\rho, u}}{2^\gamma \pi_k^\gamma / k} &= 1. \end{aligned}$$

Consequently the undirected-to-directed amplitude ratio is 2^γ for both tails and local masses, both local limiting laws have regular-variation index -2 , and no ρ -dependent factor remains.

2 The refresh-skeleton identity

Take independent sequences $Y_0, Y_1, \dots$ with common law π and $B_1, B_2, \dots$ with common law Bernoulli(q). Define

$$M_0 = 0, \quad M_n = \sum_{t=1}^n B_t, \quad X_n = Y_{M_n}.$$

This realizes exactly the stated sticky-refresh transition law and starts the chain in stationarity.

Lemma 2 (Exact skeleton identity). *For every realization and every $n \geq 0$, the sticky sample through time n and the independent word $Y_0, \dots, Y_{M_n}$ have the same vertex set and the same loop-deleted directed and undirected simple edge sets. Explicitly,*

$$\begin{aligned} V_n &= \{Y_0, \dots, Y_{M_n}\}, \\ E_n^{\rightarrow} &= \{(Y_{j-1}, Y_j) : 1 \leq j \leq M_n, Y_{j-1} \neq Y_j\}, \\ E_n^u &= \{\{Y_{j-1}, Y_j\} : 1 \leq j \leq M_n, Y_{j-1} \neq Y_j\}. \end{aligned}$$

Proof. If $B_t = 0$, then $M_t = M_{t-1}$, so the clock transition is a self-loop and is deleted. At the time of the j th refresh, $M_{t-1} = j - 1$ and $M_t = j$; the transition is therefore (Y_{j-1}, Y_j) . The refresh times up to n run once through $j = 1, \dots, M_n$. If a fresh draw repeats the present label, the transition is a self-loop on both sides and is deleted on both sides. Finally, the integer path M_t starts at zero and has increments in $\{0, 1\}$, so its attained indices through time n are exactly $0, 1, \dots, M_n$. This proves the vertex identity. Deleting loops and suppressing duplicate edges now yields both edge identities. $\square$

Let $N_n = M_n + 1$, and denote by $\hat{r}_{N,k}^a$ and $\hat{r}_{N,k+}^a$ the corresponding loop-deleted independent statistics for a word of length N . The lemma identifies all vertex degrees and the observed-range denominator. The only endpoint convention requiring comment is the directed statistic in [1], whose numerator omits a label appearing only as the final observation. If the final label appeared earlier, it is already in that numerator's domain. If it is new, it has no outgoing edge and hence out-degree zero. Thus it affects neither an exact positive-degree numerator nor a positive-degree tail numerator. For every $k \geq 1$, pathwise,

$$r_{n,k}^{\rho,a} = \hat{r}_{N,k}^a, \quad r_{n,k+}^{\rho,a} = \hat{r}_{N,k+}^a, \quad a \in \{\rightarrow, u\}. \quad (2)$$

Proof of Theorem 1. The strong law gives

$$\frac{M_n}{n} \rightarrow q > 0 \quad \text{almost surely,}$$

so $N_n \rightarrow \infty$ almost surely. For each of the four independent statistics and each positive integer k , take the probability-one event on which the fixed- k limit of Xie and Zhou holds. Their countable intersection still has probability one. On its intersection with the strong-law event, convergence of the full independent sequence implies convergence along the integer index sequence N_n . Equation (2) then gives all four clock-time limits, simultaneously for every fixed k . Independence between the random index and the limit event is unnecessary.

Substituting the four deterministic asymptotics in (1) after taking the clock-time limits proves the displayed iterated ratios. Their directed and undirected quotients give the amplitude ratio 2^γ . Finally, $\pi_k^\gamma/k \in \text{RV}_{-2}$ because $\pi_k^\gamma \in \text{RV}_{-1}$, proving the local exponent statement. $\square$

3 Scope and relation to prior work

The imported result is precisely the independent, loop-deleted theorem of Xie and Zhou [1]; the contribution here is the exact graph identity that transfers it to one dependent Markov model. No claim of literature priority is made. The argument does not exchange the n - and k -limits, cover a regime $k = k(n)$, give rates, or assert a probability-one event uniform over ρ . The condition $\rho < 1$ is essential to this reduction: at $\rho = 1$, the skeleton size is constant. Observed-range normalization is also essential to the conclusion that no sampling-rate factor remains.

For broader graph-theoretic contrast, exact deterministic structure can also transfer sharp inequalities inside restricted graph classes; an example is the induced-forest diameter bound for even-hole-free graphs in [2]. That result supplies no probabilistic or regular-variation input here. In the present setting, the entire transfer is instead furnished by the pathwise equality of two evolving simple graphs.

Computational and AI assistance. OpenAI Codex was used to help formulate and stress-test the refresh-skeleton reduction, check endpoint and normalization conventions, and prepare the manuscript. The proof itself is analytic and uses no simulation, numerical calculation, or computer-assisted lemma.

References

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